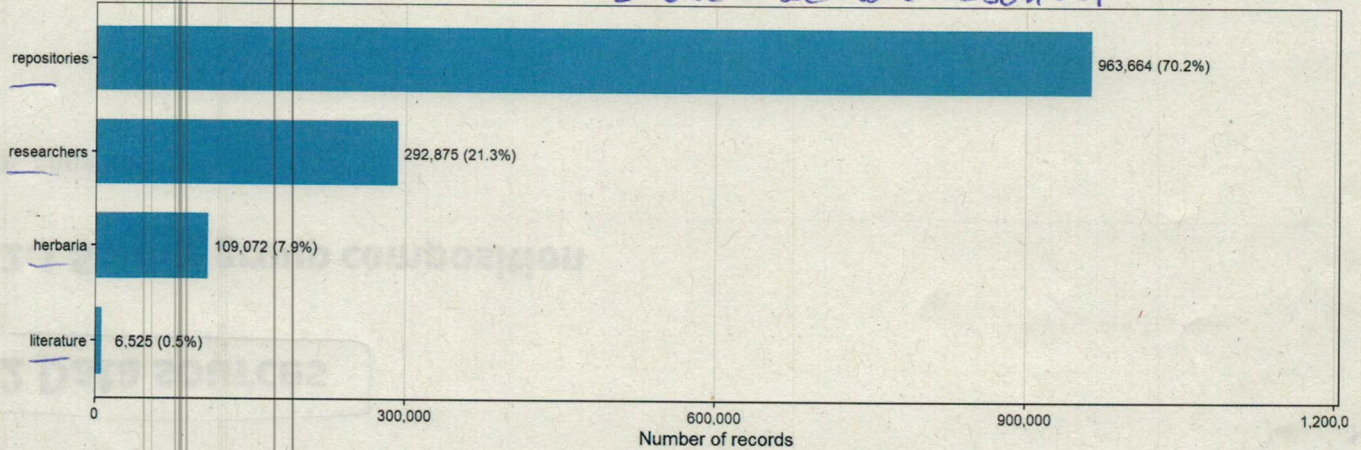


EpiG 2.0: records by source group

(this should be also described)

Records by source group. Single source groups capture the integration of multiple underlying databases per category.

3 Database overview

► Show code

EpiG 2.0 complete database: overview

Metric	Value
Total records	1,372,136 ✓ <i>full database</i>
Records with valid coordinates	1,370,421
Unique species	15,157
Unique genera	743
Unique families	107
Countries represented	53
Year range (with year information)	1692 – 2026
Median collection year	1996
Source groups	4
Databases (by source group)	herbaria: 29; literature: 2; repositories: 5; researchers: 147

overall characteristics

4 From EpiG-DB v2.0 to modelling database version

4.1 Assembly and cleaning cascade

EpiG-DB v2.0 as delivered contains 1,372,136 records: 1,263,220 from individual occurrence CSV files (herbaria + repositories) and 108,916 derived from TurboVeg community plot data (31,664 plots exported at occurrence level). This merged table has 52 columns aligned to Darwin Core and constitutes the full database product.

For the species distribution models in this thesis, a subset was prepared applying a series of conservative decisions described below. These decisions reflect the specific requirements of ensemble small-model (ESM)

Step	Exclusion	N removed	Reason
2b	EpiList filter (confirmed epiphytes only)	2,241	Cross-referenced against EpiList (confirmed Neotropical epiphyte checklist); non-epiphyte species (e.g. <i>Alpinia zerumbet</i> , <i>Cautleya spicata</i>) removed at this step
3	38 families excluded (conservative)	6,536	38 families retained in EpiG-Backbone but set aside for this thesis: the EpiList (Taylor et al. 2021) was used as reference (37 families), plus Zingiberaceae added after species-level review (<i>Alpinia</i> , <i>Cautleya</i> — Asian, not Neotropical epiphytes). Applied via <code>families_excluded.txt</code>
4	Individual genera/species excluded within retained families	7	<i>Ceiba</i> (non-epiphytic, backbone error); <i>Pyrrosia</i> (Polypodiaceae genus, native to Asia, not Neotropical)
5	Non-Neotropical records	< 600	Records outside Morrone (2022) Neotropical boundary
6	Records without valid coordinates	< 100	Records missing decimalLatitude or decimalLongitude

Families excluded via `families_excluded.txt` (38 families): These families are present in the EpiG-Backbone but were set aside for this thesis. The decision is conservative: rather than including borderline cases, families not listed in the EpiList (Taylor et al. 2021) were excluded (37 families), plus Zingiberaceae after species-level review (*Alpinia zerumbet* and *Cautleya spicata* are Asian species, not Neotropical epiphytes). *Pyrrosia* (Polypodiaceae) was excluded separately by genus filter.

► Show code

Alismataceae	Dioscoreaceae	Oxalidaceae
Alstroemeriaceae	Diplaziopsidaceae	Passifloraceae
Apiaceae	Eriocaulaceae	Petiveriaceae
Arecaceae	Fabaceae	Polygonaceae
Athyriaceae	Geraniaceae	Santalaceae
Bignoniaceae	Gleicheniaceae	Sapindaceae
Cannabaceae	Hemidictyaceae	Schoepfiaceae
Caryophyllaceae	Loganiaceae	Thelypteridaceae
Celastraceae	Loranthaceae	Verbenaceae
Convolvulaceae	Lygodiaceae	Vittariaceae
Cystopteridaceae	Marattiaceae	Woodsiaceae
Davalliaceae	Meliaceae	Zingiberaceae
Didymochlaenaceae	Menispermaceae	NA

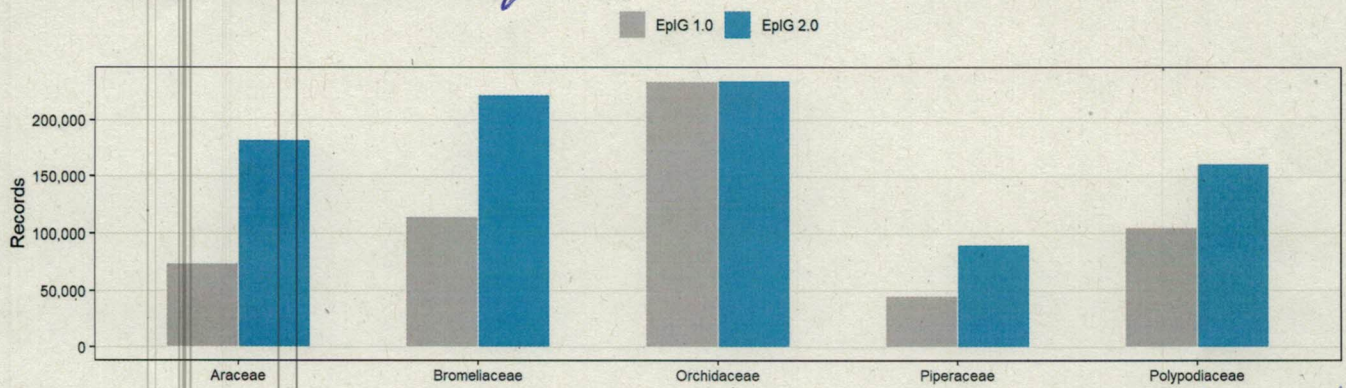
Note: The complete cleaning code is in `docs/archive/0_Join_TV_datasets.qmd`. Save run summaries to `outputs/evaluation/00_cleaning_summary.csv` for reproducibility.

5 Comparison with MSc version (included EpiG-DB v1.0)

include

preliminary version

Records per family: EpiG 1.0 vs EpiG 2.0 ✓



Record count per family: EpiG 1.0 (exact-matched records) vs EpiG 2.0.

↳ were also add/describe the rest of the family genus newly added (7 here too)

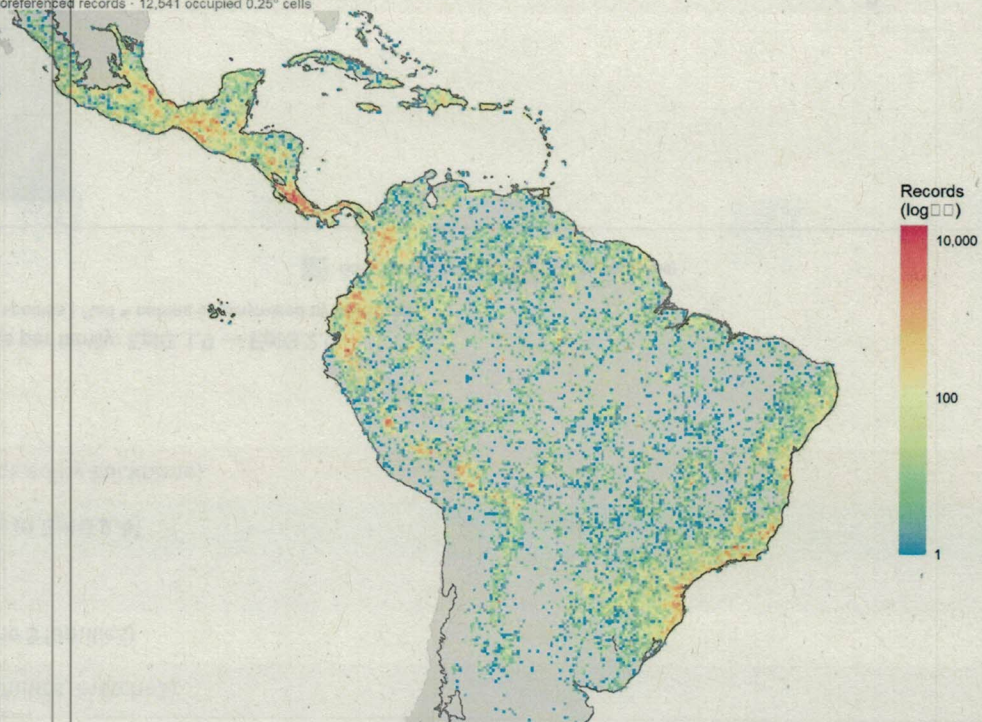
6 Geographic coverage

6.1 Occurrence density map ✓

► Show code

EpiG 2.0: occurrence density (0.25° grid)

1,370,421 georeferenced records - 12,541 occupied 0.25° cells



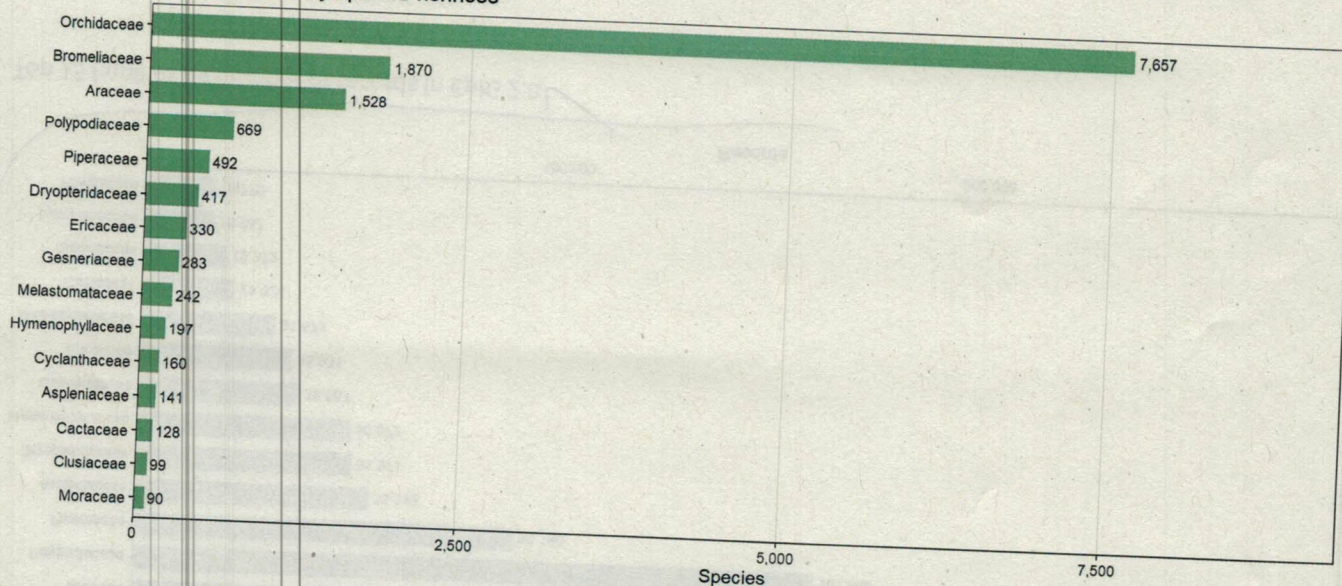
EpiG 2.0 occurrence density at 0.25° resolution on the Morrone (2022) Neotropical boundary. Red = high density; pale = low density.

6.2 Species richness map ✓

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► Show code

Top 15 families by species richness



Top 15 families by number of unique species in EpIG 2.0.

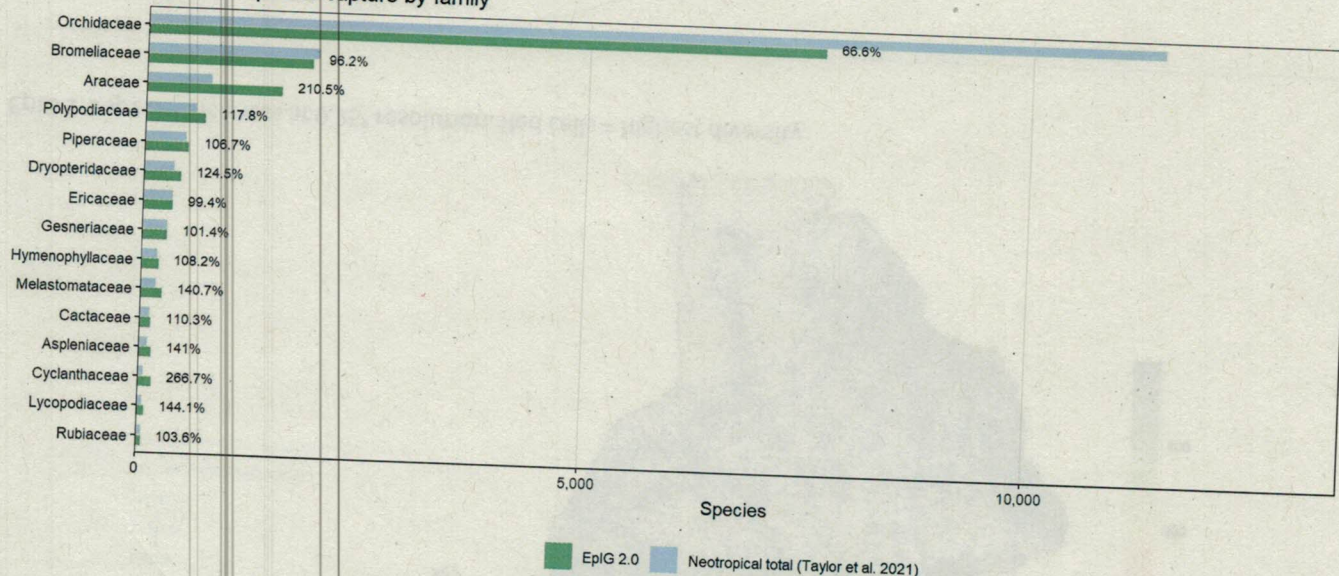
7.3 EpIG species capture vs. Neotropical total

highlight this!

How many of the epiphyte species reported for the Neotropics (Taylor et al. 2021, *Global Ecology and Biogeography*) are represented in EpIG 2.0, by family?

► Show code

EpIG 2.0 species capture by family



EpIG 2.0 species capture relative to total Neotropical epiphyte richness per family (Taylor et al. 2021). Only families present in both datasets are shown. Percentages indicate the proportion captured.

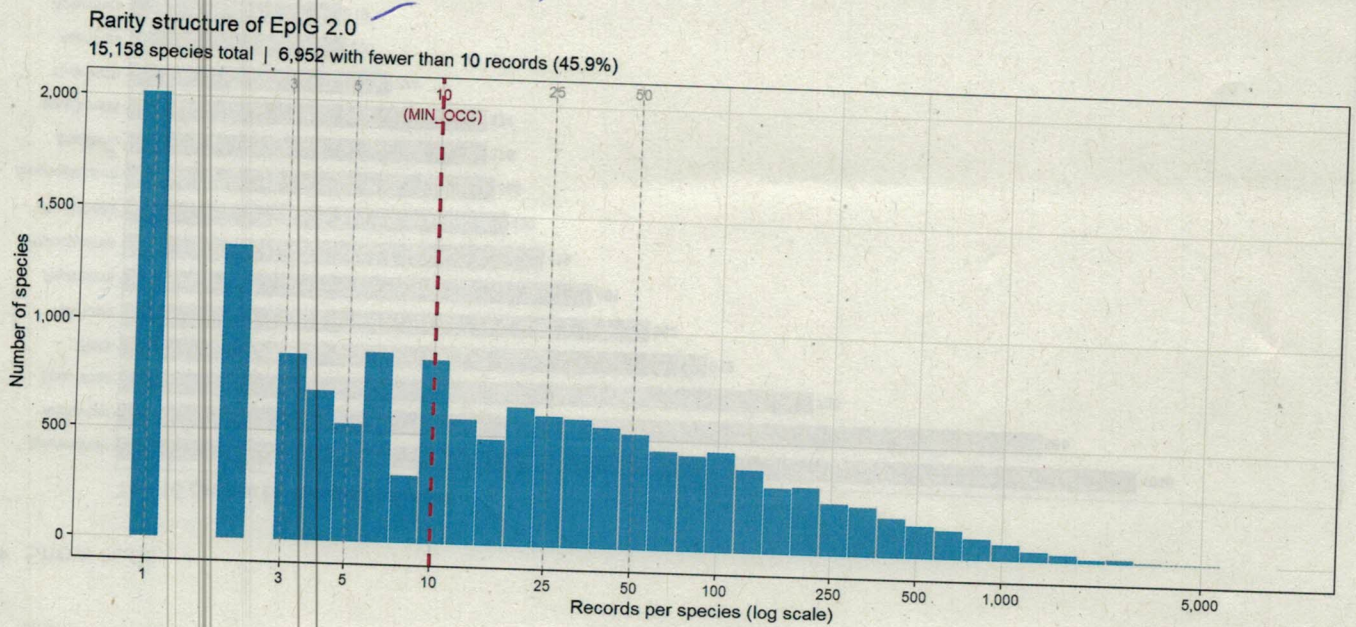
7.4 Full family summary table

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Vascular epiphytes are disproportionately rare: Svahnström et al. (2025) found that approximately 45% of epiphyte species have an extent of occurrence below 20,000 km² (the IUCN Vulnerable threshold). EpiG 2.0 reflects this pattern: the distribution of records per species is strongly right-skewed, with the majority of species known from only a handful of occurrences.

► Show code

► Show code



Number of species per record-count class (log scale). Vertical lines mark key thresholds: 1, 3, 5, 10 (ESM minimum for this study, red), 25 and 50 records.

► Show code

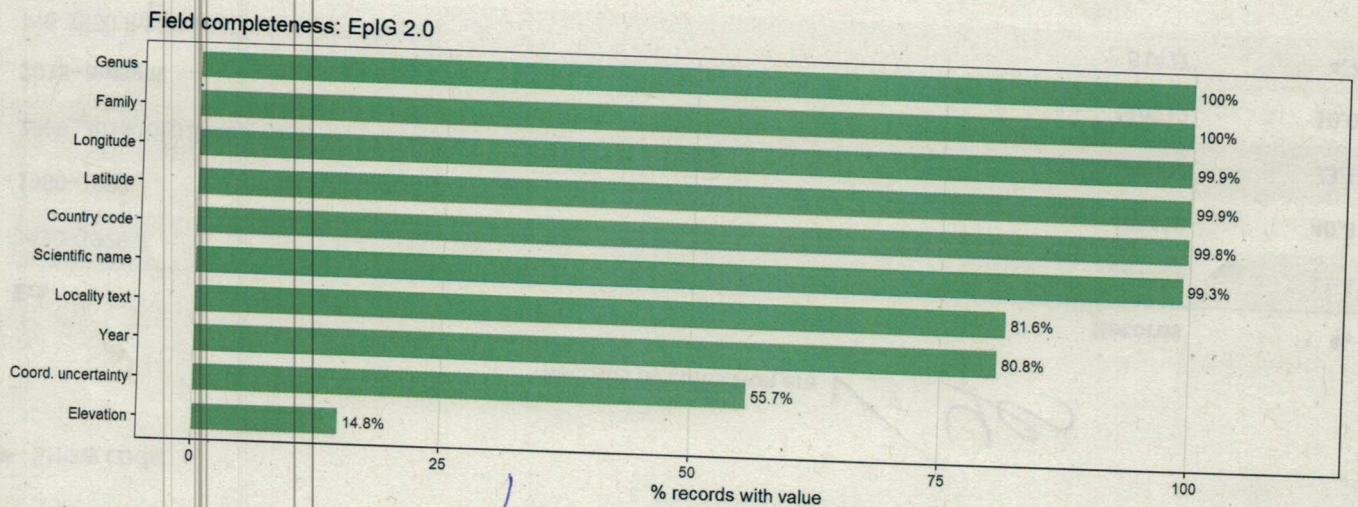
Cumulative species counts at key record-count thresholds. Each row shows how many species have at least that many records in EpiG 2.0 (pre-thinning).

Threshold (n ≥)	N species	% of all spp	Note
1	15157	100.0	Any record
3	11798	77.8	Reference cut
5	10244	67.6	Old ESM minimum (Breiner 2015 lower bound)
10	8205	54.1	MIN_OCC: ESM threshold for this study
25	5736	37.8	Reference cut
50	3997	26.4	Moderate data: well-estimated niches
100	2581	17.0	High data: standard SDM feasible
250	1214	8.0	Very high: rare in endemic epiphytes

► Show code

10 Field completeness

► Show code



Completeness of key fields in EpiG 2.0.

Not a graph but table?

the same as
in community data